Unified 4D World Action Modeling from Video Priors with Asynchronous Denoising

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Research Questions

- How can unified world action models incorporate explicit 3D spatial awareness without sacrificing computational efficiency or pretrained visual priors?
- How can joint video generation and action prediction be optimized for both high-fidelity video synthesis and real-time action execution?
- Can a single unified framework simultaneously excel at policy execution, 4D world modeling, and spatial reconstruction?

Core Idea

Existing unified world action models are limited to 2D pixel space, lacking 3D geometric understanding crucial for robotics. X-WAM addresses this by adding a lightweight depth prediction branch that replicates the final blocks of a pretrained video diffusion model, enabling 4D world modeling without doubling sequence length. The key insight is Asynchronous Noise Sampling (ANS): during training, it samples joint noise distributions where video timesteps are always >= action timesteps, matching inference where actions are decoded quickly (few steps) for real-time execution while videos continue denoising for high quality.

Key Formula

Asynchronous Noise Sampling (ANS)

```
(t_O, t_a) \sim \begin{cases} t_a = 0, t_O \sim U(0, 1) & \text{w.p. } p \\ t_a \sim U(0, 1), t_O = t_a + (1 - t_a) \cdot b, b \sim \text{Beta}(1.5, 1) & \text{w.p. } 1-p \end{cases}
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Joint noise sampling where t_O (video timestep) and t_a (action timestep) are coupled. First case samples action-conditioned video generation with noise-free actions. Second case ensures t_O >= t_a via Beta distribution, matching inference where actions decode faster than video, eliminating training-inference mismatch.

Key Figure

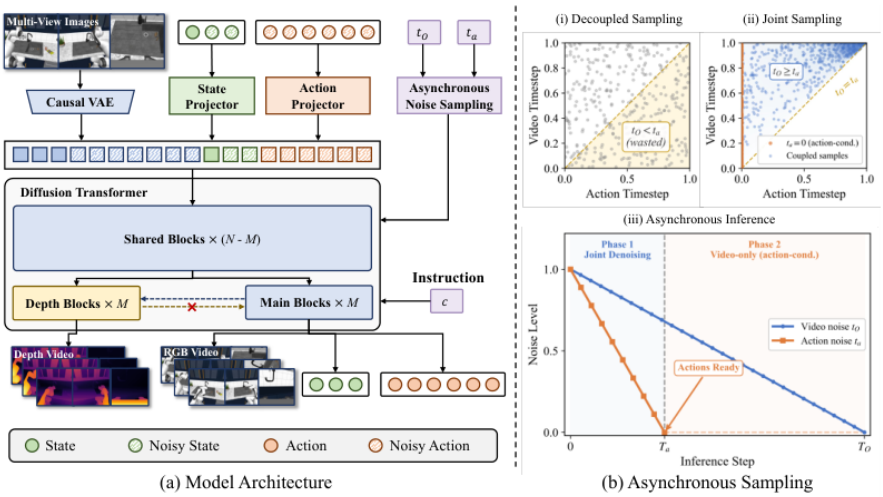

Figure 2: **Overview of X-WAM.** (a) Model architecture: multi-view RGB observations, proprioceptive states, and noisy actions are encoded and jointly denoised by a Diffusion Transformer initialized from Wan2.2-5B, with a lightweight interleaved depth branch for spatial modeling. (b) Asynchronous Noise Sampling (ANS): (i) standard decoupled sampling wastes training on configurations where $t_O < t_a$; (ii) our coupled joint sampling ensures $t_O \geq t_a$, faithfully matching the inference distribution; (iii) during inference, actions are decoded in T_a steps and immediately dispatched, while video denoising continues for T_O steps.

Figure 2: Shows the core architectural design with lightweight depth branch and illustrates how ANS couples training noise distributions to match asynchronous inference behavior.

Limitations

- Requires multi-view camera setup which may not be available in all robotic systems
- Depth supervision relies on pretrained depth estimation models during training, inheriting their limitations
- Asynchronous inference still requires Ta=10 steps for action decoding, limiting maximum control frequency
- Model size and computational requirements may be prohibitive for edge deployment on resource-constrained robots